| Article | Imaging modality | Database | Scope | Setting, data source and outcome | Sample Size(train, (validation) ,test) | Machine learning technique | Performance (best) (%) | Validation type | ROB |
|---|---|---|---|---|---|---|---|---|---|
| [27] | Chest X-ray | - | Segmentation | early diagnosis and longitudinal follow-up of suspected pneumonia | 22k(unclear) | Deep learning | - | unclear | High |
| [28] | CT | - | Classification and segmentation | evaluate an artificial intelligence (AI) system in differentiating COVID-19 and other pneumonia on chest CT and assess radiologist performance without and with AI assistance | 32583 (830,237,119 ) | Deep learning(CNN) | Accuracy: 96 Sensitivity: 95 Specificity: 96 | External | High |
| [30] | CT | data from China and US | Classification | present a system that utilizes robust 2D and 3D deep learning models | 50 patient (unclear) | Deep learning | C index 99 (98 to 100) | External | High |
| [32] | CT | Tongji Hospital of Huazhong University | Classification and segmentation | propose to conduct diagnosis of COVID-19 with multi-view representation learning | 2522 (70%, 30%)) | SVM, Logistic-Regression (LR), Gaussian- Naive-Bayes (GNB), K-Nearest-Neighbors (KNN), And Fully-Connected-Neural Networks (NN) | Accuracy: 93 Sensitivity: 100 Specificity: 91 | 5-fold cross validation | High |
| [34] | CT | data from China, US, and Switzer-land | Classification | propose an artificial intelligence (AI) system for fast COVID-19 diagnosis | 416 patient (220,196) | Deep learning | C index 98, Sensitivity: 94, Specificity: 95 | Training test split | High |
| [73] | CT | data from Italy | Classification | early phase detection of Coronavirus (COVID-19) by machine learning methods | 53 (unclear) | GLCM, LDP, GLRLM, GLSZM, DWT and SVM | Sensitivity: 93, Specificity: 100 | Cross validation | High |
| [35] | CXR | Nonpublic+kgg dataset | Classification | evaluated an AI system for detection of COVID-19 characteristics on frontal chest radiographs | 24678 (23138,1540, 468) | Deep learning | AUC: 81 Sensitivity:85 Specificity: 65 | Training test split | High |
| [36] | CXR | Dr.Joseph Cohen data+ dataset | Classification | introduced a transfer of deep learning models to classify COVID-19 X-ray images | 307 (reach 8100 after GAN) | Deep learning | Accuracy: 100 Binary Accuracy: 84-class Google Net has had | Training test split | High |

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| | | Chest X-ray Images | | | (80%, 20%) | | the best performance for 4-class scenario -> Precision: 84, F1 Score:82 ,Recall: 80 | | |
| [37] | CT | Iran University of medical science | Classification | propose a CAD system based on deep learning to classify COVID-19 infection versus other a typical and viral pneumonia dis- eases | 1020 CT (80%, 20%) | Deep learning | Accuracy: 99 Sensitivity: 100 Specificity:99 | Training test split | unclear |
| [38] | Chest X-ray | COVID Chest X-ray ,Kaggle and pneumonia | Classification | rapid diagnostic of the COVID-19 based on deep Bayes-Squeeze Net | 5949 (80%, 10%, 10%) | Deep learning | Accuracy: 98, Specificity: 99, F1: 98 | Training test split | High |
| [39] | Chest X-ray | Chest X-ray scans | Classification | developed a CNN-based model to perform COVID-19 detections from chest radiographers | 108k (80%, 10%, 10%) | Deep learning | Accuracy: 96 ,Precision: 91, Recall: 97 | Training test split | High |
| [40] | X-ray and CT | chest X-ray and CT dataset | Multiclass classification | training and classification on several deep model | 6087 (80%, 20%) | Deep learning | Accuracy: 92 Sensitivity: 92 Specificity: 96 Precision: 92 F1:0.92 | Training test split | High |
| [41] | CT | CC-CCII | Segmentation and Classification | developed an AI system that can diagnose NCP and differentiate it from other common pneumonia and normal controls | 617775 (80%, 10%, 10%) | Deep learning and Decision tree(GBDT) | Accuracy: 96 , Dice: 95 | 5-fold cross validation | High |
| [42] | CT | - | Classification | trained a multi-view fusion model using deep learning network to screen patients with COVID-19 using CT image | 495 (395,50,50) | Deep learning | Accuracy: 76 Sensitivity:81 Specicificity:61 | Training test split | High |

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| [76] | X-ray | COVID-19, SARS, Pneumocystis and Streptococcus Images Varicella And MERS images | Classification | novel proposed hierarchical classification approach | 3049 (2696,353) | K-Nearest Neighbors (KNN); Support Vectors Machine (SVM); Multilayer Perceptron (MLP); Decision Trees (DT); and Random Forests (RF) | F1 Score: 89 | 5-fold cross validation | High |
| [43] | CT | COVID-19 dataset + CT-EGFR | Classification and segmentation | proposed a novel fully automatic DL system using raw chest CT image to help COVID-19 diagnostic and prognostic analysis | 4106 + 1266 (709,4 validation group(226, 161,53, 117) | Deep learning | Accuracy:85, Sensitivity:80, Specificity:89, F1Score:90 | External | High |
| [45] | CT | Nonpublic dataset(COVID 19 and CAP) | Classification | proposed a novel fully automatic DL system using raw chest CT image to help COVID-19 diagnostic and prognostic analysis | 497 + 498(CA) (490, 82, 77) | Deep learning | Sensitivity:84 Specificity:76 AUC: 70 | Training test split | High |
| [46] | X-ray | NIH Chest X-Ray dataset | Classification | appropriately localized abnormalities for both focal and multi focal disease | 11000+88(frontal CXR (unclear) | Deep learning | Accuracy: 89 | External | High |
| [75] | X-ray | dataset from Kaggle and GitHub | Classification | presented an automated corona detection method by using novel feature generation model and an iterative feature selector | 321 lung X-ray images (unclear) | Relief (IRF) based feature selection ,Decision tree linear discriminant (LD), support vector machine (SVM), k nearest neighbor- hood (KNN) and | Accuracy: 99 Sensitivity: 98 Specificity: 100 | 10-fold cross validation | High |

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| | | | | | | subspace discriminant (SD) | | | |
| [74] | CT | Beijing Youan hospital | Segmentation and clustering | lung lesion segmentation ,image preprocessing, feature extraction, feature selection and signature construction | 75 Patients (50, 25) | SVM and Radiometric feature extraction | AUC: 73 | 3-fold cross validation | Unclear |
| [47] | CXR | GitHub source with MERS, SARS, COVID-19 and Qatar University | Classification and feature extraction + transfer learning | Preprocessed each image with deep learning models+ data reconstructed using the Fuzzy technique + training by deep learning models + classified by SVM | 458 images (70%, 30%) | SVM and deep learning | Accuracy: 98 Sensitivity: 99 Specificity:99 | 5-fold cross validation | High |
| [48] | CXR | chest x-ray8 database | Classification | Deep learning model is proposed for the automatic diagnosis of COVID-19 | 1127 (80%, 20%) | Deep learning | Accuracy( Binary): 98 Accuracy: Multi:87 Sensitivity:85 Specificity: 92 Precision: 89 F1 Score: 87 | 5-fold cross validation | High |
| [49] | CTR | Nonpublic dataset | Classification | proposed a three-dimensional deep learning | 4356 chest CT (90%, 10%) | Deep learning | AUC: 96 Sensitivity: 90 Specificity: 96 CAP -> AUC: 95 Sensitivity: 87 Specificity: 92 non-pneumonia -> AUC: 98 Sensitivity: 94 Specificity: 96 | Training test split | High |
| [77] | CT | Chest CT images dataset | Classification | multi-objective differential evolution (MODE)–based convolutional neural net- works(CNN) for classification of COVID-19-infected patients from chest CT images | Unclear (Different validation ratio) | Deep learning | ACC: 1.9789, F-measure:2.0928 Sensitivity: 1.8262 Specificity:1.6827 Kappa statistics:1.9276 | 20-fold cross validation | High |

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| [50] | CT | Nonpublic dataset | Classification | Used artificial intelligence (AI) algorithms to rapidly diagnose patients who are positive for COVID-19 | 905 patients (80%, 20%) | Deep learning | Accuracy: 17.25<br>AUC: 92<br>Sensitivity: 84<br>Specificity: 82 | Training test split | High |
| [51] | CXR | COVID-19 X-Ray | Classification | state-of-the-art Convolutional Neural Network called Mobile Net is employed and trained from scratch to investigate the importance of the extracted features for the classification task | 3905 X-ray images (90%, 10%) | Deep learning | Accuracy: 91<br>Sensitivity: 97<br>Specificity: 99<br>Accuracy<br>7-class: 51 | 10-fold cross validation | High |
| [52] | CT | - | Classification + CAM | develop a DL model to evaluate the efficacy of Dense Net for detection of COVID-19 features on high resolution CT | 295 (135,20,140) | Deep learning | AUC: 98<br>Accuracy: 92<br>Sensitivity: 97<br>Specificity: 87<br>F1 Score: 93 | four-fold cross validation | High |
| [53] | CXR | COVID-19 radiography database kaggle dataset + data synthesis using GAN | Classification | Auxiliary Classifier (AC- GAN) COVID GAN for the generation of synthetic CXR images.+ Design a CNN-based Using COVID GAN for augmentation of training dataset | 1124 (932, 192) | Deep learning | Accuracy: 95<br>Sensitivity: 90<br>Specificity: 97 | Training test split | High |
| [54] | CT | TCIA dataset | Classification and segmentation | presented a novel weakly supervised deep learning framework that is capable of learning to detect and localize lesions on COVID- 19 and CAP CT scans from image-level label only | 60 3D CT (40, 10, 10) | Deep learning | Accuracy: 90<br>AUC: 95<br>Sensitivity: 90<br>Specificity: 91 | 5-fold cross validation | High |
| [55] | CXR | JSRT Dataset + NLM dataset | Classification and segmentation | Propose a patch-based convolutional neural network approach with a relatively small number of trainable parameter for COVID-19 diagnosis. | 247 JSRT 138 USNLM (354, 49, 99) | Deep learning | Accuracy: 88<br>Precision: 83<br>Recall: 85<br>F1 Score: 84<br>Specificity:96 | External | High |

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| [56] | CTR | - | Classification | propose an attention based deep 3D multiple instance learning (AD3D- MIL) | 460 chest CT (60%, 20%, 20%) | Deep learning | Binary: Accuracy:97 AUC: 99 F1 Score: 97 Precision:97 Recall:97 Multi-Class: Accuracy: 94 AUC: 98 F1 Score: 92 Precision:95 Recall:90 | 5-fold cross validation | High |
| [57] | CT | LUNA-16 dataset+ nonpublic dataset | Classification + CAM | lobe segmentation using 3D-Unet and)pneumonia prediction using 3D-ResNets with prior- attention mechanism | 4597 (3997,600) | Deep learning | Accuracy: 93 Sensitivity: 87 Specificity: 95 | 5-fold cross validation | High |
| [58] | LUS (lung ultra-sonography) | Italian COVID-19 Lung Ultrasound Database (ICLUS-DB) | Classification and segmentation | present a novel fully-annotated dataset of LUS images collected from several Italian hospitals, with labels indicating the degree of disease severity at a frame-level, video level, and pixel-level (segmentation masks). | 58924 Frames (70%, 30%) | Deep learning | Accuracy: 96 DICE:75 binary DICE:65 multi-class | 5-fold cross validation | High |
| [60] | CT | 100 labeled CT slices From the COVID-19 CT Segmentation dataset And 1600 unlabeled images from the | Classification and segmentation | Lung Infection Segmentation Deep Network (Inf-Net) is proposed to automatically identify infected regions from chest CT slices + semi-supervised segmentation framework based on a randomly selected propagation strategy, | 1700 Labeled: (45, 5, 50) | Deep learning | MAE: 6.4 | Training test split | High |

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| | | COVID-19 CT Collection dataset | | | | | | | |
| [61] | CT | data from China | Classification | construct a system based on deep learning for detecting COVID-19 pneumonia on high resolution CT | 46096 anonymous images ( 40, 11)patients | Deep learning | Sensitivity:100 Specificity: 82 | Training test split | High |
| [63] | CT | data from China | Classification | building and deploying an AI system that automatically analyzes CT images to detect COVID-19 pneumonia features | 1136 patient (723, 259,154) | Deep learning | C index: 99, Sensitivity: 97 Specificity:92 | Training test split | High |
| [64] | CT | data from China | Segmentation | a deep learning (DL)-based segmentation system is developed to automatically quantify infection regions of interest (ROIs) and their volumetric ratios w.r.t. the lung | 249 patient (unclear) | Deep learning | Dice similarity coefficient 91.6 | Training test split | High |
| [65] | CT | data from China | Classification and segmentation | All images were preprocessed to obtain the segmentations of both infections and lung fields, which were used to extract location-specific features | 2685 patient (80%, 20%) | Random Forest | C index 94 | 5-fold cross validation | High |
| [66] | CT | data from China | Classification | using deep learning methods to extract COVID-19's specific graphical features and provide a clinical diagnosis | 338 patient (259, 79) | Deep learning | C index 81 (71 to 84), Sensitivity: 83, Specificity 67 | Internal, other image from same people | High |
| [67] | CT | data from China | Classification | establish an early screening model to distinguish COVID-19 pneumonia from Influenza-A viral pneumonia and healthy cases with pulmonary CT images using deep learning techniques | 629 patient (509,90, 30) | Deep learning and Bayesian function | Sensitivity 87 PPV:81 | Training test split | High |

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|---|---|---|---|---|---|---|---|---|---|
| [69] | CT | data from China | Classification | develop an accurate computer-aided method to assist clinicians to identify COVID-19-infected patients by CT images | 192 patient (131,34,27) | Deep learning | C index 96 | Training test split | High |
| [70] | CT | data from China | Classification | Developing a deep learning-based model for automatic COVID-19 detection on chest CT | unclear | Deep learning | C index 96 | Temporal validation | High |
| [71] | CT | data from China | Prediction | establish an AI model to predict mild patients with potential malignant progression | 187 patients (133,54) | Deep learning and logistic regression | C index 95 (94 to 97) | 5-fold cross validation | High |
| [72] | CT | data from China | Prediction | Intervention CT radiomics models based on logistic regression (LR) and random forest (RF) were developed on features extracted from pneumonia lesions in training and inter-validation datasets | 31patients (26,5) | logistic regression (LR) and random forest (RF) | C index 92 | 5-fold cross validation | High |